Images and evidence should be included throughout.

Manager should make a copy, share with each member of the group and (jmiranda@rpsk12.org, pnguyen@rpsk12.org), make it viewable by "ANYONE with the link" and submit it to Google Classroom when complete. Delete the notes in red when you are done with each section.

Group Members: poeople

Element D: Design Concept Generation, Analysis and Selection

Possible Solutions

Solution Name	Solution Description (include sketches)	Pros	Cons	Inspiration for idea
Basketball	Using rubber bands and 3D printed platforms to put the metal balls to determine the slope	It can have greater range and more varieties of strength	The rubber band can break	I asked some of my friends like
Archery	By simply using a rubber band and 2 toothe pick your are able to make a bow and arrow then using the 3D printer we are able to make a moveable target (has to be manually moved) depending how far back or close the target is the arrow will move in a downwards direction	• Students will be more interested • Game will have a point system which will make the game more competitive • Students will be able to see the slope depending how far or close the target is	• Very Difficult to hit a perfect bullseye • Near impossible to hold the bow and arrow • Immature students might point it at classmates ending up with someone injured • Students might lose pieces to the game	I used to be a professional archer
Volleyball	Similar to the basketball game but instead its a two player using all the same components such as the moveable platform, 3D printed ball platform the main difference is that there will be a net	• Students will be more engaged • Students will be able to play against eachother • Students will be able to change/move around the platforms • Can also be used by one person • Can also be used as an exampled during math class	• If one of the components break it will be difficult to make • If something goe missing it will be difficult to play	I used to play volleyball

Process	This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Words and drawings/photos should be included. Each possible solution needs to include distinguishing details such as scale, dimensions, and materials. Talk about improvements/changes you made to the solutions.

Decision Matrix

Ratings (how well solution addresses requirement) 0 = not at all, 1 = very little, 2 = somewhat, 3 = mostly, 4 = completely

Design Requirements		Design Solutions (Score/Weighted Score)			
Requirement	Weight (add up to 100)	Solution 1….	Solution 2…	Solution 3…	Solution 4…
Weighted Total Score					

Design Decision

Detailed description of chosen design	

Blueprint or sketches of your design	Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs
Justification	
Detailed Plan	A Gawntt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc. For a level 4, the plan of action should be detailed enough that someone could repeat the same steps and testing for effectiveness like a cookbook or recipe.

	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
D1. Process for generating & comparing possible solutions. The process for generating possible solutions ___________.	was comprehensive, iterative, and consistently defensible (viable design highly likely)	was thorough, iterative, and generally defensible (viable design likely)	was adequate, generally iterative, and generally defensible (viable design is possible)	was partial or overly general and only somewhat iterative and/or defensible (raising issues with the design viability)	was incomplete or minimally iterative and/or defensible (raising issues with the design viability)	was not evident that there was an attempt
D2. Design solution choice. The design solution chosen ___________.	was well justified and demonstrated attention to all design	was justified and demonstrated attention to most if not all design	was explained with reference to at least some design requirements	was not sufficiently explained with reference to design requirements	lacked support related to design requirements	was not attempted or evidence not provided

	requirements	requirements				
D3. Plan of action. The plan of action __________.	has considerable merit and would easily support repetition and testing for effectiveness by others	would support repetition and testing for effectiveness by others	might not clearly or fully support repetition and testing for effectiveness by others	has insufficient detail to allow for testing for replication of results	is lacking to the point that the design solution cannot be tested	cannot be executed or does not exist
D4. Design blueprint. Attributes of the design solution chosen are presented __________.	in a consistently specific, detailed, and thorough manner	in a generally specific, detailed, and thorough manner	in a somewhat specific and detailed manner, but may not be thorough	in a somewhat specific and/or detailed, but incomplete or overly general manner	in a sparse manner, with few specifics/ details	minimally